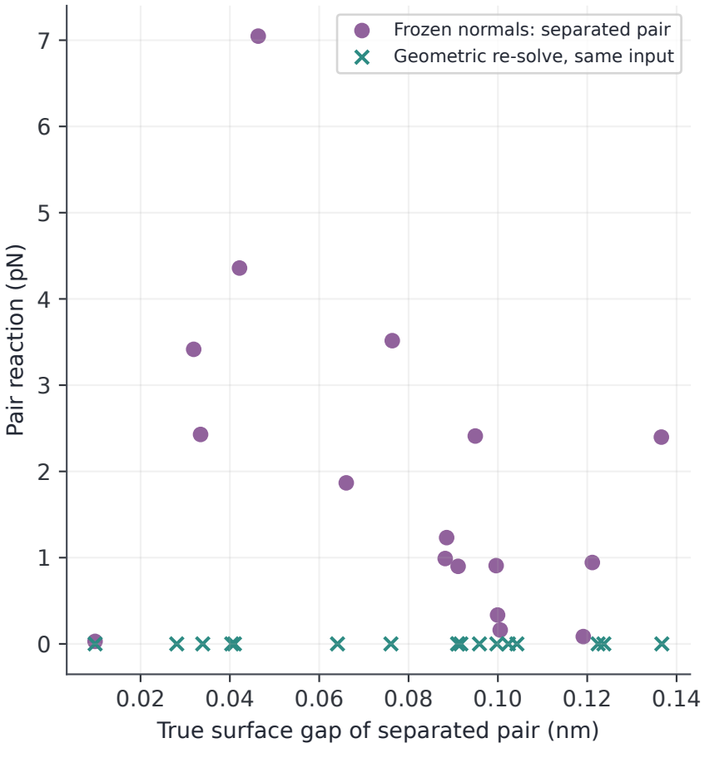
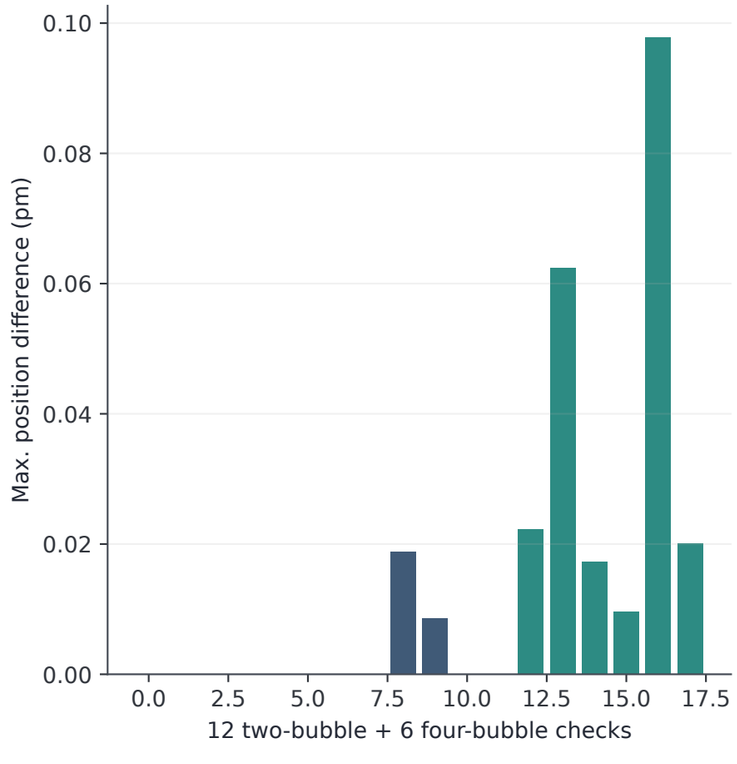
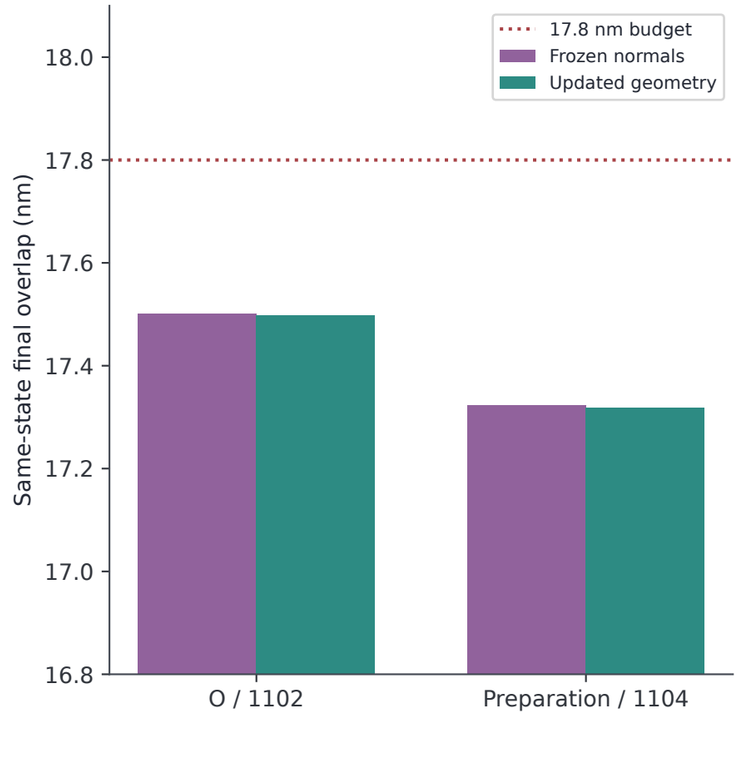
a Archived separated-pair reactions



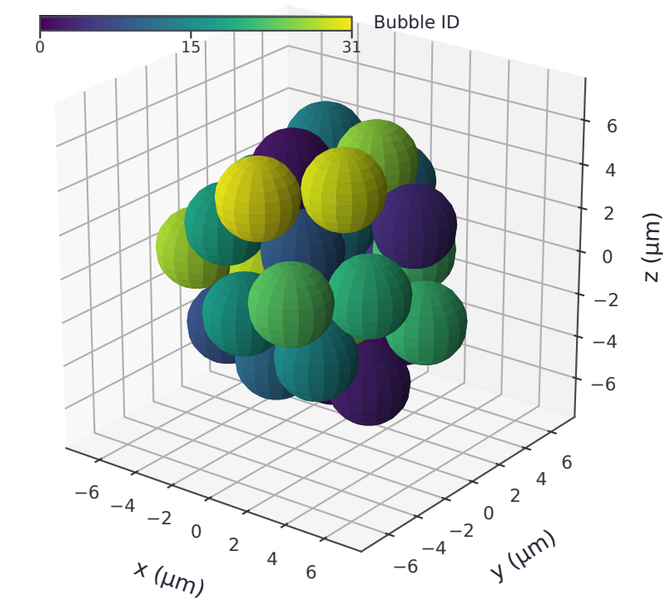
b Independent nonlinear reference



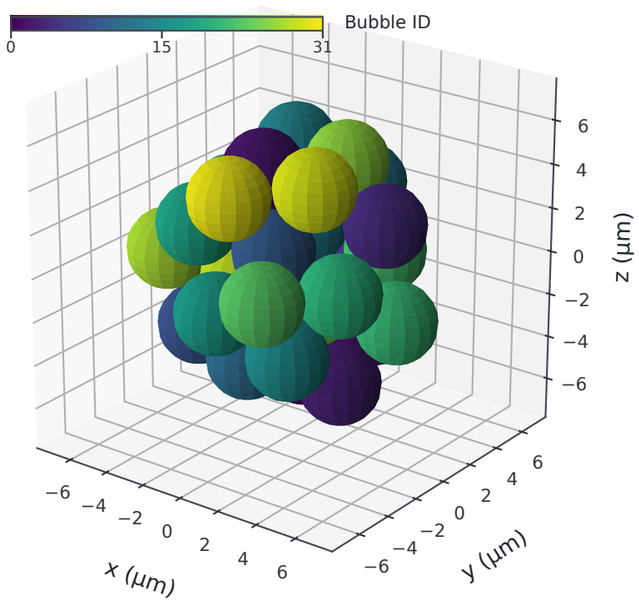
c Saved32 static states, fixed 2kc



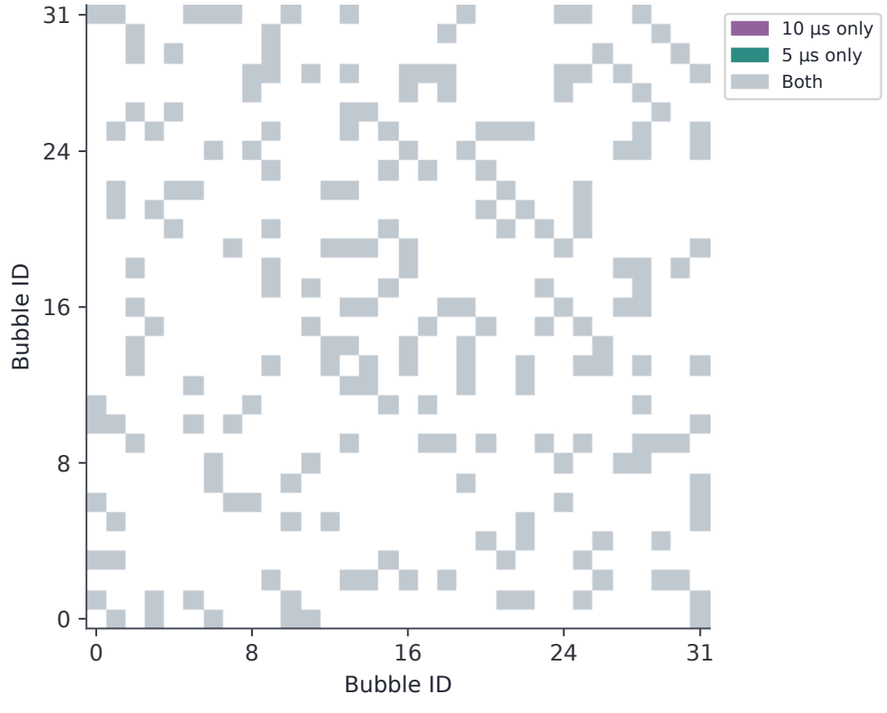
d Geometric B2 / 10 µs, t = 5 ms



e Geometric B2 / 5 µs, t = 5 ms



f Fixed-ID connectivity at 5 ms



Fixed kc = 0.0464132 N/m; an uncalibrated numerical candidate. Geometric spring reactions use current distance and normal.
True radius = 1.78 µm; equal XYZ scales and fixed ID colors. Connections use d <= 2.1R; matrix preserves every fixed-ID edge.
Seventeen same-input pair-event re-solves and evolved 5 ms states are distinct evidence. No sphere shrinkage, smoothing or fabricated trajectory.